Angles in Triangles and Quadrilaterals

Name _____ Date _____ No calculator Version 1

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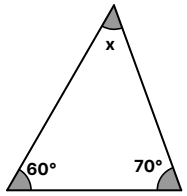
KEY WORDS triangle *(tapatoru)* · isosceles *(tapatoru waewae-rite)* · equilateral *(tapatoru rite)*

A Angles in a triangle ●○○

Remember: The three angles inside any triangle add to 180°.

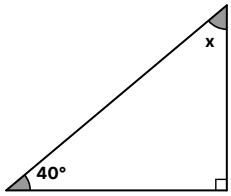
1. Find the size of the angle marked x.

a)



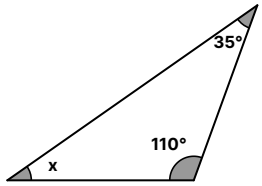
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b)



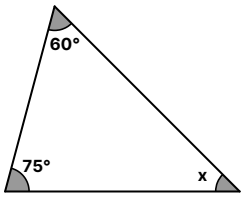
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c)



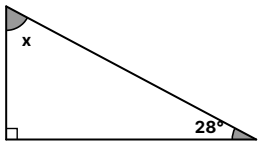
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d)



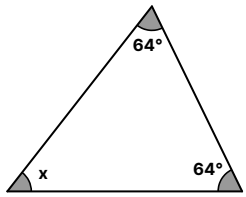
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e)



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f)



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2. Answer each question.

a) Two angles of a triangle are 35° and 80°. Find the third angle. _____

b) Two angles of a triangle are 90° and 47°. Find the third angle. _____

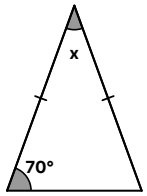
c) Two angles of a triangle are 118° and 26°. Find the third angle. _____

B Isosceles and equilateral triangles ●●○

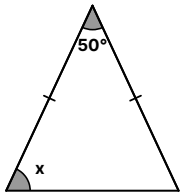
Remember: An isosceles triangle has two equal sides (marked with small lines) and two equal angles opposite them. An equilateral triangle has three equal sides, and every angle is 60°.

3. Find the size of the angle marked x. Give a reason.

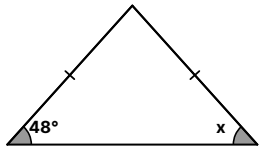
a)



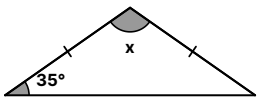
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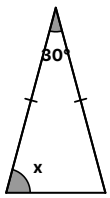
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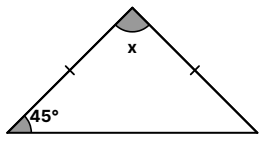
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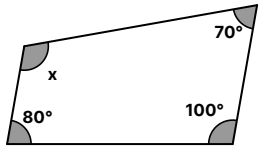
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C Angles in a quadrilateral ●●○

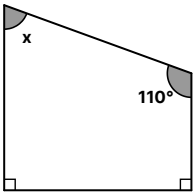
Remember: The four angles inside any quadrilateral add to 360°.

4. Find the size of the angle marked x.

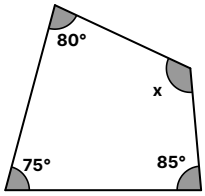
a)



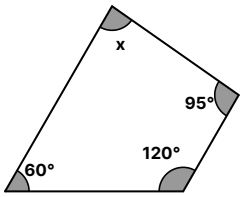
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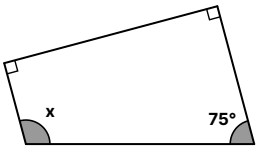
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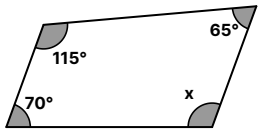
d)



e)



f)



5. Answer each question.

a) Three angles of a quadrilateral are 90°, 85° and 110°. Find the fourth angle. _____

b) Three angles of a quadrilateral are 72°, 108° and 64°. Find the fourth angle. _____

c) Three angles of a quadrilateral are 135°, 90° and 45°. Find the fourth angle. _____

D Using algebra ●●●

Remember: Add the angles, put the total equal to 180 (triangle) or 360 (quadrilateral), then solve the equation.

6. Write an equation and solve it to find x.

a) The angles of a triangle are x°, $2x^\circ$ and $3x^\circ$.

b) The angles of a triangle are x°, $x + 20^\circ$ and $x + 40^\circ$.

c) The angles of a quadrilateral are x°, x°, $2x^\circ$ and $2x^\circ$.

d) The angles of a quadrilateral are x°, $2x^\circ$, $x + 30^\circ$ and $x + 80^\circ$.

★ Extension

7. One angle of a right-angled triangle is 35°. Find the size of its third angle.

8. Can a triangle have two obtuse angles? Explain.

9. Two angles of a triangle are 50° and 60°. One side is made longer at the third corner. Find the angle between this extended side and the triangle (the exterior angle).

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Angles in Triangles and Quadrilaterals

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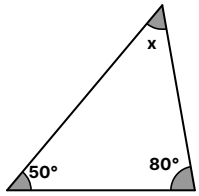
KEY WORDS triangle *(tapatoru)* · isosceles *(tapatoru waewae-rite)* · equilateral *(tapatoru rite)*

A Angles in a triangle ●○○

Remember: The three angles inside any triangle add to 180°.

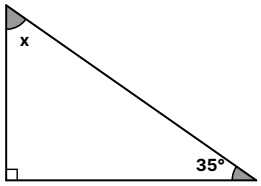
1. Find the size of the angle marked x.

a)



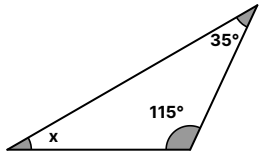
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b)



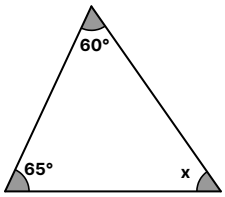
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c)



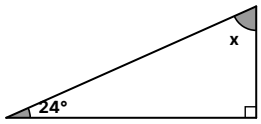
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d)



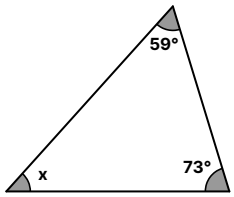
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e)



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_____

2. Answer each question.

a) Two angles of a triangle are 45° and 70°. Find the third angle. _____

b) Two angles of a triangle are 90° and 53°. Find the third angle. _____

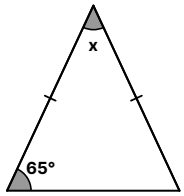
c) Two angles of a triangle are 124° and 19°. Find the third angle. _____

B Isosceles and equilateral triangles ●●○

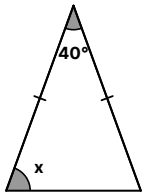
Remember: An isosceles triangle has two equal sides (marked with small lines) and two equal angles opposite them. An equilateral triangle has three equal sides, and every angle is 60°.

3. Find the size of the angle marked x. Give a reason.

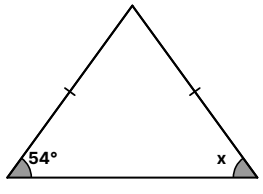
a)



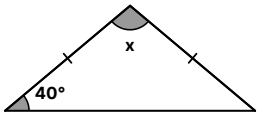
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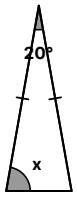
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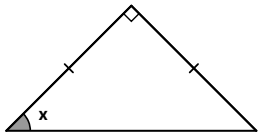
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C Angles in a quadrilateral ●●○

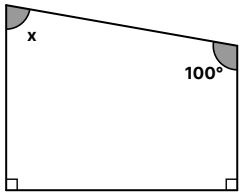
Remember: The four angles inside any quadrilateral add to 360°.

4. Find the size of the angle marked x.

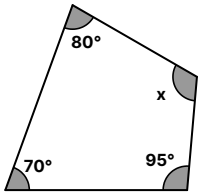
a)



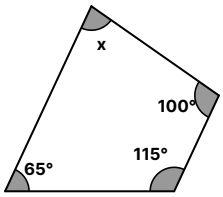
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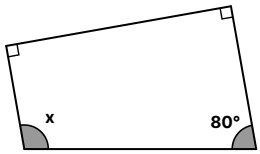
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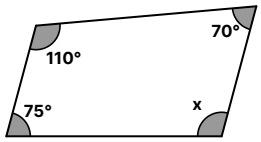
d)



e)



f)



5. Answer each question.

a) Three angles of a quadrilateral are 90°, 95° and 105°. Find the fourth angle. _____

b) Three angles of a quadrilateral are 68°, 112° and 74°. Find the fourth angle. _____

c) Three angles of a quadrilateral are 140°, 90° and 50°. Find the fourth angle. _____

D Using algebra ●●●

Remember: Add the angles, put the total equal to 180 (triangle) or 360 (quadrilateral), then solve the equation.

6. Write an equation and solve it to find x.

a) The angles of a triangle are x°, $3x^\circ$ and $5x^\circ$.

b) The angles of a triangle are x°, $x + 10^\circ$ and $x + 50^\circ$.

c) The angles of a quadrilateral are x°, $2x^\circ$, $3x^\circ$ and $3x^\circ$.

d) The angles of a quadrilateral are x°, $2x^\circ$, $x + 40^\circ$ and $x + 70^\circ$.

★ Extension

7. One angle of a right-angled triangle is 28°. Find the size of its third angle.

8. Can a triangle have two obtuse angles? Explain.

9. Two angles of a triangle are 45° and 75°. One side is made longer at the third corner. Find the angle between this extended side and the triangle (the exterior angle).

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